

Decentralised Task Allocation under Uncertainty

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1 Introduction

This project explores task allocation strategies for a fleet of vehicles operating in a dynamic environment with uncertainty. Tasks arrive over time with varying urgency and duration, where urgency increases every step, and vehicles must be assigned efficiently while considering constraints such as battery level, travel time, and availability.

The goal of this work is to compare different allocation strategies and evaluate their performance under identical simulation conditions. In particular, a reinforcement learning approach is implemented alongside heuristic baselines, i.e. Greedy based allocation and Auction based allocation to understand its effectiveness in this setting.

2 Methodology

Three allocation strategies were implemented and compared:

1. A greedy approach, which assigns each task to the most suitable available vehicle based on immediate cost.
2. An auction-based method, where vehicles bid for tasks based on their local state.
3. A Q-learning-based approach, where an agent learns task allocation policies through interaction with the environment.

The environment is dynamic, with new tasks arriving probabilistically at each time step. Task urgency increases over time if not serviced or given attention, introducing additional pressure on the system.

3 Implementation

The system was implemented using a discrete-time simulation. At each time step:

1. Vehicle states are updated (busy/idle status and remaining duration)
2. Task urgency is incremented
3. New tasks may be generated
4. Allocation is performed using each strategy independently

For Q-learning, the state representation includes task urgency, distance to the vehicle, battery level, and system load. The reward function balances urgency, travel time, engagement time, and remaining battery.

To ensure robust evaluation, each experiment was repeated multiple times and average metrics were computed.

4 Results

The performance of each method was evaluated using the following metrics:

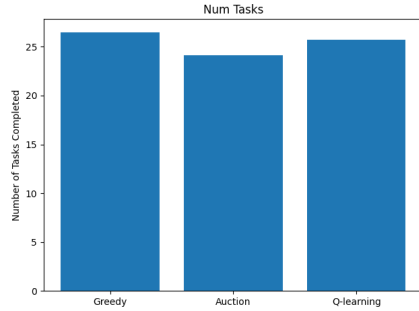
1. Number of tasks completed [Figure 1a]
2. Throughput (total task duration serviced) [Figure 1b]
3. Average idle time [Figure 1c]
4. Energy consumption per unit task [Figure 1d]

The greedy approach consistently achieved the highest throughput and task completion. The Q-learning approach performed competitively, outperforming the auction method in most cases but slightly underperforming compared to the greedy baseline.

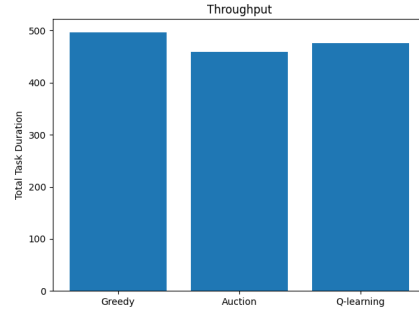
5 Discussion

The results indicate that in this relatively structured environment, the greedy approach performs near-optimally due to its strong focus on immediate efficiency.

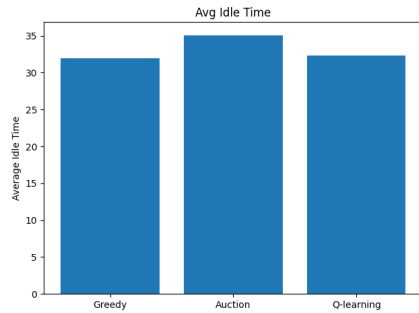
While the Q-learning model is capable of learning allocation strategies, its advantage is limited by the lack of strong long-term dependencies in the



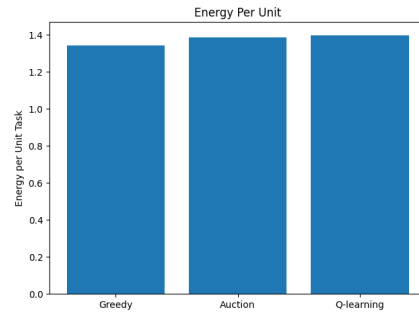
(a) Tasks Completed



(b) Throughput



(c) Average Idle Time



(d) Energy Efficiency

Figure 1: Performance comparison of Greedy, Auction, and Q-learning approaches

environment. Since most decisions depend on immediate factors, the learned policy converges to behaviour similar to the greedy approach.

However, the reinforcement learning framework remains valuable due to its adaptability and potential to handle more complex or uncertain environments.

6 Conclusion

This study compared heuristic and learning-based approaches for decentralised task allocation. While the greedy strategy achieved the best performance in the current setup, the Q-learning approach demonstrated competitive results and offers a foundation for handling more complex scenarios.

Future work could explore richer state representations and environments

with stronger temporal dependencies to better leverage the strengths of reinforcement learning.